Graph analysis guide

High-Impact Teaching Math Tools Without the Price Tag

Start with the question

What is being measured? Who or what was sampled? Is the question about categories, change over time, a distribution, or a relationship?

Check the display

Read the title, source, axis labels, units, intervals, and baseline. Look for omitted categories, unequal intervals, distorted areas, and missing context. Is the representation appropriate for the question?

Garden challenge

Synthetic data: Garden A = 12 insects; Garden B = 15 insects. Graph A has a vertical baseline of 0. Graph B has a vertical baseline of 10. Which exaggerates the visual difference?

Answer and reasoning

Graph B. The actual difference is 3 insects; 15 is 25% more than 12. The visible lengths above ten are 2 and 5, a ratio of 2.5. A zero baseline preserves bar-length comparisons of totals. Some line graphs can reasonably use nonzero baselines; always explain the context.

Student explanation prompts

My claim is ____. The axis and values show ____. My calculation is ____. A limitation of this dataset is ____. I would need more information about ____.

Sampling matters

Equal-area sampling and equal observation effort matter. Two isolated counts do not describe variation or establish why the locations differ. Collect repeated observations before drawing broader conclusions.

Choose a representation

Categories: bar chart. Ordered change: line graph. Parts exhausting a whole: pie chart. Paired numerical variables: scatter plot. Distribution: dot plot, histogram, or box plot as appropriate. The question and structure of the data determine the choice.

Source and further learning

https://splatworks.org/solutions/graphsplat/lesson-plans

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